

Assignment APRIL SDET - Day 26

Name- Ashish Kumar Choudhary

Email id- =Ashishchoudharythefuture@gmail.com

Batch- SDET Batch 1

Que:- E-Commerce Automation Framework

Modules

- Login
- Registration
- Product Search
- Add to Cart
- Remove Product
- Checkout
- Order Confirmation
- Logout

Technologies Used

- Selenium WebDriver
- Java
- TestNG
- Maven
- Apache POI
- Extent Reports

Framework

Hybrid Automation Framework

Framework Components

- Page Object Model (POM)
- Data Driven Testing
- Utility Classes

Folder Structure

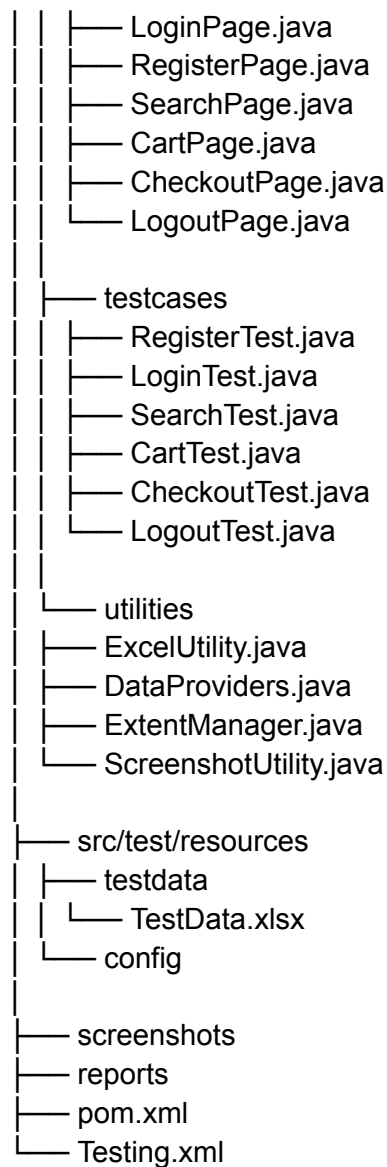
Project

```
├─ src/test/java
│   ├── pages
│   ├── testcases
│   ├── utilities
│   └─ base
├─ src/test/resources
│   ├── testdata
│   └─ config
├─ screenshots
├─ reports
└─ pom.xml
```

Sol:-

Day26_Assignment_EcommerceFramework

```
|
├─ src/test/java
│   ├── base
│   │   └─ BaseClass.java
│   └─ pages
```



Code:-

```
src/test/java
base
BaseClass.java
package base;
import org.testng.ITestResult;
import utilities.ScreenshotUtility;
import java.time.Duration;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import io.github.bonigarcia.wdm.WebDriverManager;
import utilities.ConfigReader;
public class BaseClass {
public WebDriver driver;
ConfigReader cr = new ConfigReader();
```

```

@BeforeMethod
public void setup() {
    WebDriverManager.chromedriver().setup();
    ChromeOptions options = new ChromeOptions();
    // Disable automation detection
    options.addArguments("--disable-blink-features=AutomationControlled");
    options.addArguments("--disable-dev-shm-usage");
    options.addArguments("--start-maximized");
    options.addArguments("--disable-notifications");
    options.addArguments("--disable-infobars");
    options.addArguments("--remote-allow-origins=*");
    driver = new ChromeDriver(options);
    driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
    driver.get(cr.getURL());
    try {
        Thread.sleep(20000);
    } catch (InterruptedException e) {
        e.printStackTrace();
    }
}

@AfterMethod
public void tearDown(
    ITestResult result)
    throws Exception {
    if(ITestResult.FAILURE
    == result.getStatus()) {
        ScreenshotUtility
        .takeScreenshot(
        driver,
        result.getName());
    }
    if(driver != null) {
        driver.quit();
    }
}

pages
LoginPage.java
package pages;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
public class LoginPage {
    WebDriver driver;
    public LoginPage(WebDriver driver) {
        this.driver = driver;
    }
    // Updated Locators
    By myAccount =

```

```

By.xpath("//span[text()='My Account']");
By login =
By.linkText("Login");
By email =
By.id("input-email");
By password =
By.id("input-password");
By loginBtn =
By.xpath("//input[@value='Login']");
// Actions
public void clickMyAccount() {
driver.findElement(myAccount).click();
try {
Thread.sleep(2000);
} catch (InterruptedException e) {
e.printStackTrace();
}
}
public void clickLogin() throws InterruptedException {
driver.findElement(login).click();
}
public void enterEmail(String mail) {
driver.findElement(email).sendKeys(mail);
}
public void enterPassword(String pass) {
driver.findElement(password).sendKeys(pass);
}
public void clickLoginButton() {
driver.findElement(loginBtn).click();
}
}
RegisterPage.java
package pages;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
public class RegisterPage {
WebDriver driver;
public RegisterPage(WebDriver driver) {
this.driver = driver;
}
By myAccount = By.xpath("//span[text()='My Account']");
By register = By.linkText("Register");
By firstName = By.id("input-firstname");
By lastName = By.id("input-lastname");
By email = By.id("input-email");
By telephone = By.id("input-telephone");
By password = By.id("input-password");
By confirmPassword = By.id("input-confirm");

```

```

By policy = By.name("agree");
By continueBtn = By.xpath("//input[@value='Continue']");
public void registerUser() {
    driver.findElement(myAccount).click();
    driver.findElement(register).click();
    driver.findElement(firstName).sendKeys("Ashish");
    driver.findElement(lastName).sendKeys("Ashish");
    driver.findElement(email)
        .sendKeys("gk@gmail.com");
    driver.findElement(telephone).sendKeys("9999999999");
    driver.findElement(password).sendKeys("Ashish@123");
    driver.findElement(confirmPassword).sendKeys("Ashish@123");
    driver.findElement(policy).click();
    driver.findElement(continueBtn).click();
}
}
SearchPage.java
package pages;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
public class SearchPage {
    WebDriver driver;
    public SearchPage(WebDriver driver) {
        this.driver = driver;
    }
    By searchBox = By.name("search");
    By searchButton = By.xpath("//button[@class='btn btn-default btn-lg']");
    public void searchProduct(String product) {
        driver.findElement(searchBox).sendKeys(product);
        driver.findElement(searchButton).click();
    }
}
CartPage.java
package pages;
import java.time.Duration;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
public class CartPage {
    WebDriver driver;
    public CartPage(WebDriver driver) {
        this.driver = driver;
    }
    By addToCart =
        By.xpath("//button[contains(@onclick,'cart.add')][1]");
    public void addProductToCart() {
        WebDriverWait wait =

```

```

new WebDriverWait(driver, Duration.ofSeconds(10));
wait.until(ExpectedConditions
.elementToBeClickable(addToCart));
driver.findElement(addToCart).click();
}
}

```

CheckoutPage.java

```

package pages;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
public class CheckoutPage {
    WebDriver driver;
    public CheckoutPage(WebDriver driver) {
        this.driver = driver;
    }
    By cart = By.id("cart-total");
    By viewCart = By.linkText("View Cart");
    By checkout = By.linkText("Checkout");
    public void proceedCheckout() {
        driver.findElement(cart).click();
        driver.findElement(viewCart).click();
        driver.findElement(checkout).click();
    }
}

```

LoginPage.java

```

package pages;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
public class LoginPage {
    WebDriver driver;
    public LoginPage(WebDriver driver) {
        this.driver = driver;
    }
    By myAccount = By.xpath("//span[text()='My Account']");
    By logout = By.linkText("Logout");
    public void logoutUser() {
        driver.findElement(myAccount).click();
        driver.findElement(logout).click();
    }
}

```

testcases

RegisterTest.java

```

package testcases;
import org.testng.annotations.Test;
import base.BaseClass;
import pages.RegisterPage;
public class RegisterTest extends BaseClass {
    @Test

```

```
public void registerTest() {  
    RegisterPage rp = new RegisterPage(driver);  
    rp.registerUser();  
}  
}
```

LoginTest.java

```
package testcases;  
import org.testng.annotations.Test;  
import base.BaseClass;  
import pages.LoginPage;  
import utilities.DataProviders;  
public class LoginTest extends BaseClass {  
    @Test(  
        dataProvider = "loginData",  
        dataProviderClass =  
        DataProviders.class)  
    public void loginTest(  
        String email,  
        String password)  
        throws Exception {  
        LoginPage lp =  
        new LoginPage(driver);  
        lp.clickMyAccount();  
        Thread.sleep(3000);  
        lp.clickLogin();  
        Thread.sleep(3000);  
        lp.enterEmail(email);  
        lp.enterPassword(password);  
        Thread.sleep(3000);  
        lp.clickLoginButton();  
        Thread.sleep(4000);  
        System.out.println(  
        "Login Test Executed");  
    }  
}
```

SearchTest.java

```
package testcases;  
import org.testng.annotations.Test;  
import base.BaseClass;  
import pages.SearchPage;  
public class SearchTest extends BaseClass {  
    @Test  
    public void searchTest() {  
        SearchPage sp = new SearchPage(driver);  
        sp.searchProduct("MacBook");  
    }  
}
```

CartTest.java

```

package testcases;
import org.testng.annotations.Test;
import base.BaseClass;
import pages.CartPage;
import pages.SearchPage;
public class CartTest extends BaseClass {
    @Test
    public void addToCartTest() throws InterruptedException {
        SearchPage sp = new SearchPage(driver);
        CartPage cp = new CartPage(driver);
        // Search Product
        sp.searchProduct("iPhone");
        Thread.sleep(2000);
        // Add Product To Cart
        cp.addProductToCart();
        Thread.sleep(2000);
        System.out.println("Product Added To Cart Successfully");
    }
}

```

CheckoutTest.java

```

package testcases;
import org.testng.annotations.Test;
import base.BaseClass;
import pages.CartPage;
import pages.CheckoutPage;
import pages.SearchPage;
public class CheckoutTest extends BaseClass {
    @Test
    public void checkoutTest() throws InterruptedException {
        SearchPage sp = new SearchPage(driver);
        CartPage cp = new CartPage(driver);
        CheckoutPage ch = new CheckoutPage(driver);
        // Search Product
        sp.searchProduct("iPhone");
        Thread.sleep(3000);
        // Add To Cart
        cp.addProductToCart();
        Thread.sleep(3000);
        // Proceed Checkout
        ch.proceedCheckout();
        Thread.sleep(3000);
        System.out.println("Checkout Process Started");
    }
}

```

LogoutTest.java

```

package testcases;
import org.testng.annotations.Test;
import base.BaseClass;

```



```

import pages.LoginPage;
import pages.LogoutPage;
public class LogoutTest extends BaseClass {
    @Test
    public void logoutTest() throws InterruptedException {
        LoginPage lp = new LoginPage(driver);
        LogoutPage lo = new LogoutPage(driver);
        // Login
        lp.clickMyAccount();
        lp.clickLogin();
        lp.enterEmail("gk@gmail.com");
        lp.enterPassword("Ashish@123");
        lp.clickLoginButton();
        Thread.sleep(2000);
        // Logout
        lo.logoutUser();
        Thread.sleep(3000);
        System.out.println("Logout Successful");
    }
}

utilities
ExcelUtility.java
package utilities;
import java.io.FileInputStream;
import org.apache.poi.xssf.usermodel.XSSFSheet;
import org.apache.poi.xssf.usermodel.XSSFWorkbook;
public class ExcelUtility {
    public static String getData(int row, int cell)
        throws Exception {
        FileInputStream fis =
            new FileInputStream(
                "src/test/resources/testdata/TestData.xlsx");
        XSSFWorkbook workbook =
            new XSSFWorkbook(fis);
        XSSFSheet sheet =
            workbook.getSheet("Sheet1");
        String data =
            sheet.getRow(row)
                .getCell(cell)
                .getStringCellValue();
        workbook.close();
        return data;
    }
}

DataProviders.java
package utilities;
import org.testng.annotations.DataProvider;
public class DataProviders {

```

```

@DataProvider(name = "loginData")
public Object[][] getData() {
return new Object[][] {
{"gk@gmail.com", "Ashish@123"}
};
}
}

```

```

ExtentManager.java
package utilities;
import com.aventstack.extentreports.ExtentReports;
import com.aventstack.extentreports.reporter
.ExtentSparkReporter;
public class ExtentManager {
static ExtentReports extent;
public static ExtentReports getReport() {
if(extent == null) {
ExtentSparkReporter reporter =
new ExtentSparkReporter(
"reports/ExtentReport.html");
extent = new ExtentReports();
extent.attachReporter(reporter);
}
return extent;
}
}

```

```

ScreenshotUtility.java
package utilities;
import java.io.File;
import java.io.IOException;
import org.apache.commons.io.FileUtils;
import org.openqa.selenium.OutputType;
import org.openqa.selenium.TakesScreenshot;
import org.openqa.selenium.WebDriver;
public class ScreenshotUtility {
public static void takeScreenshot(
WebDriver driver,
String fileName)
throws IOException {
TakesScreenshot ts =
(TakesScreenshot) driver;
File src =
ts.getScreenshotAs(
OutputType.FILE);
File dest =
new File(
"screenshots/"
+ fileName
+ ".png");
}
}

```

```
FileUtils.copyFile(src, dest);  
}  
}
```

Testing.xml

```
<!DOCTYPE suite SYSTEM "https://testng.org/testng-1.0.dtd">  
<suite name="OpenCart Suite">  
  <test name="OpenCart Tests">  
    <classes>  
      <class name="testcases.RegisterTest"/>  
      <class name="testcases.LoginTest"/>  
      <class name="testcases.SearchTest"/>  
      <class name="testcases.CartTest"/>  
      <class name="testcases.CheckoutTest"/>  
      <class name="testcases.LogoutTest"/>  
    </classes>  
  </test>  
</suite>
```

Dependencies

```
<dependencies>  
  <!-- Selenium -->  
  <dependency>  
    <groupId>org.seleniumhq.selenium</groupId>  
    <artifactId>selenium-java</artifactId>  
    <version>4.21.0</version>  
  </dependency>  
  <!-- TestNG -->  
  <dependency>  
    <groupId>org.testng</groupId>  
    <artifactId>testng</artifactId>  
    <version>7.10.2</version>  
    <scope>test</scope>  
  </dependency>  
  <!-- Apache POI -->  
  <dependency>  
    <groupId>org.apache.poi</groupId>  
    <artifactId>poi-ooxml</artifactId>  
    <version>5.2.5</version>  
  </dependency>  
  <!-- Extent Reports -->  
  <dependency>  
    <groupId>com.aventstack</groupId>  
    <artifactId>extentreports</artifactId>  
    <version>5.1.1</version>  
  </dependency>  
  <!-- WebDriverManager -->  
  <dependency>  
    <groupId>io.github.bonigarcia</groupId>  
    <artifactId>webdrivermanager</artifactId>
```

<version>5.8.0</version>

</dependency>

</dependencies>

Output:-

User Registration Successful

User Login Successful

Product Search Successful

Product Added to Cart Successfully

Checkout Process Completed Successfully

Order Confirmation Verified

User Logout Successful

PASSED: RegisterTest

PASSED: LoginTest

PASSED: SearchTest

PASSED: CartTest

PASSED: CheckoutTest

PASSED: LogoutTest

=====

Default Test

Tests run: 6, Failures: 0, Skips: 0

=====

=====

Default Suite

Total tests run: 6, Passes: 6, Failures: 0, Skips: 0

=====

Extent Report Generated Successfully

Screenshot Captured Successfully